

DE Lab11
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Q1:

a)

$$V_{TN} = V_{T0} + \gamma(\sqrt{(1.2 - V_{TN}) + 2|\phi_F|} - \sqrt{2|\phi_F|})$$

$$V_{TN} = 0.4 + 0.2(\sqrt{(1.2 - V_{TN}) + 0.88} - \sqrt{0.88})$$

$$V_{TN} = 0.46644V$$

$$A = B = C = 1.2 - V_{TN} = 0.73356V$$

$$V_{TN2} = V_{T0} + \gamma(\sqrt{(A - V_{TN2}) + 2|\phi_F|} - \sqrt{2|\phi_F|})$$

$$V_{TN2} = 0.42997V$$

$$D = E = F = A - V_{TN2} = 0.73356 - 0.42997 = 0.30359V$$

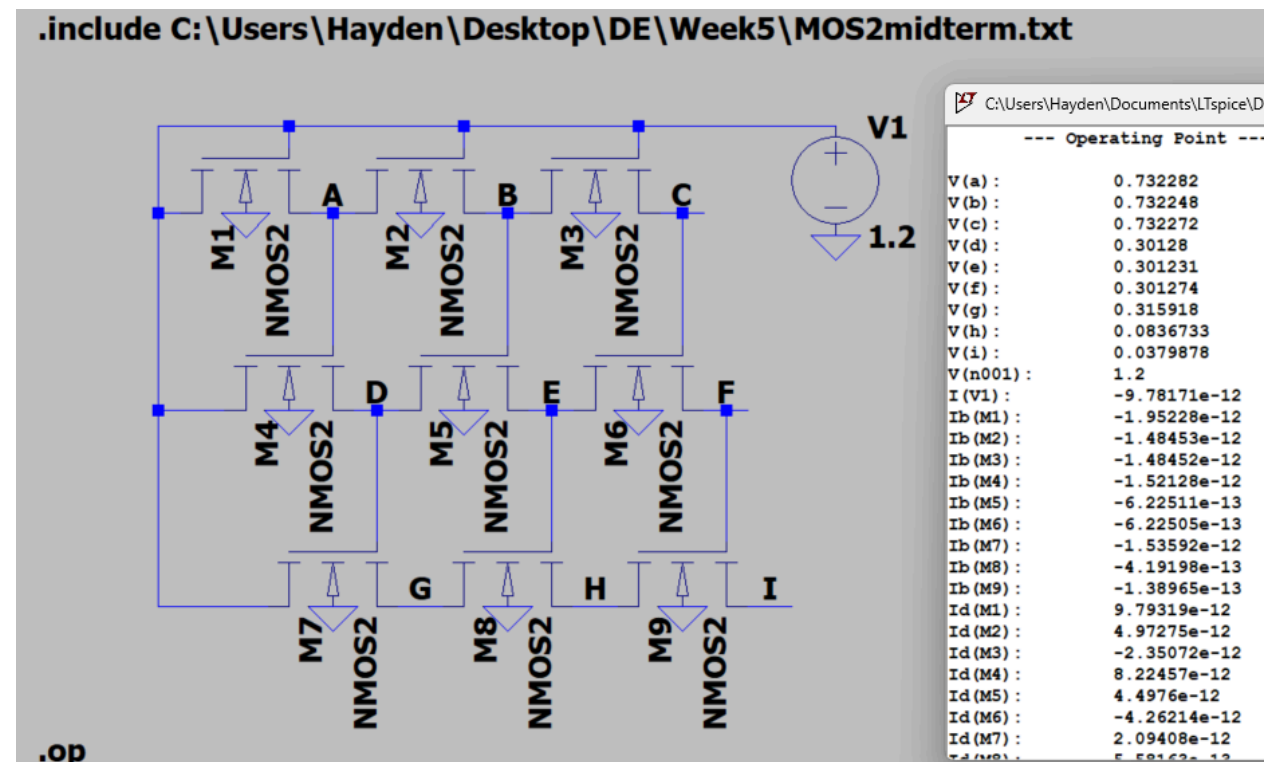
$$V_{TN3} = V_{T0} + \gamma(\sqrt{(D - V_{TN3}) + 2|\phi_F|} - \sqrt{2|\phi_F|})$$

$$V_{TN3} = 0.3905V$$

$$D < V_{TN3}$$

$$G = H = I = 0V$$

b)



Values all match up fairly well, G=0.316V is the only one significantly higher than calculated

Q2:

when a rises (first time), F and X “average out”, $1.2 \cdot 10f + 0 \cdot 2f = V_F \cdot 10f + V_X \cdot 2f$
 limited by $V_X = V_F - V_{TN}$

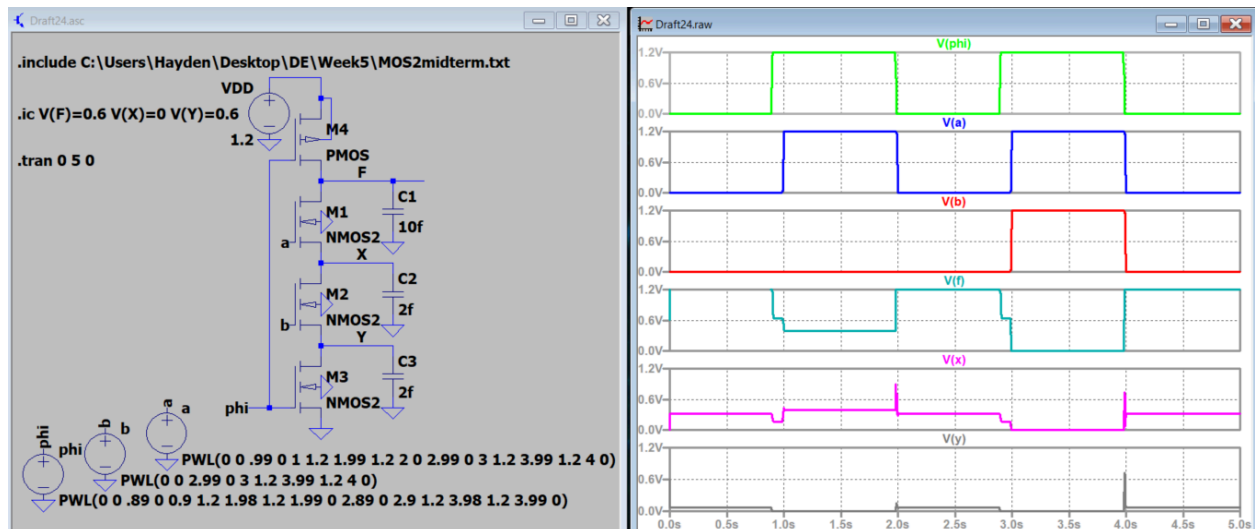
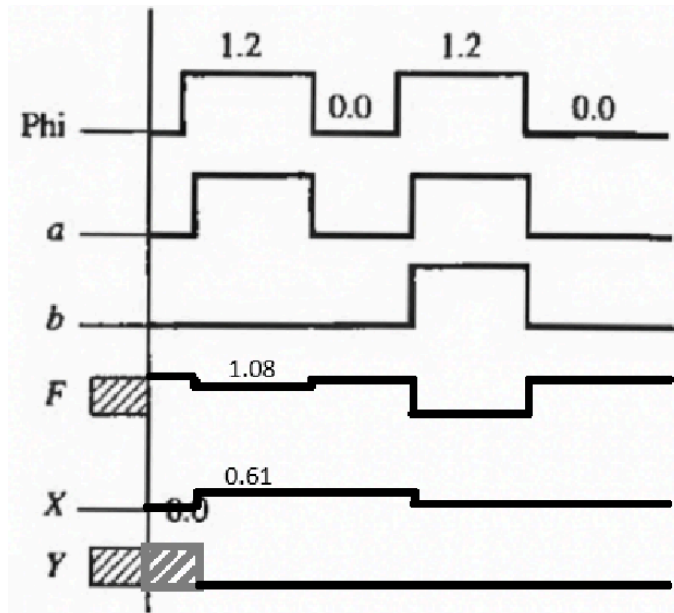
$V_{TN} = V_{T0} + \gamma(\sqrt{(1.2 - V_{TN}) + 2|\phi_F|} - \sqrt{2|\phi_F|}) = 0.46644V$

$12 = 10 \cdot V_F + 2 \cdot V_X$, $V_X = V_F - 0.46644V$

$V_X = 6 - 5 \cdot V_F$

$V_F = 1.07774V$

$V_X = 0.6113V$



F drops more than calculated but all plots follow the right trends